Multidisciplinary Action Project Report

On

NEW PRODUCT DEVELOPMENT OF PROBIOTIC CURD AND IMPROVEMENT OF PACKAGING AT VIJAY DAIRY

Submitted to Institute Code: 750

S. R. Luthra Institute of Management

Under the Guidance of

Dr. Roshni Singh

Assistant Professor

In partial Fulfilment of the Requirement of the award of the degree of

Master of Business Administration (MBA)

Offered By

Gujarat Technological University

Ahmadabad

Prepared by:

NAME	ENROLLMENT NUMBER
Mahipalsinh A. Magodra	207500592036
Avani P. Kachhadiya	207500592038
Isha D. Gorasiya	207500592046
Shivani B. Jasoliya	207500592056
Saurabhkumar B. Kansagara	207500592085

MBA (Semester - III)

Month & Year: JANUARY 2022

Students' Declaration

We hereby declare that the **Multidisciplinary Action Project** titled **"New Product Development and Improvement of Packaging at Vijay Dairy" is** a result of our own work and our indebtedness to other work publications, references, if any, have been duly acknowledged. If we are found guilty of copying from any other report or published information and showing as our original work, or extending plagiarism limit, we understand that we shall be liable and punishable by the university, which may include being declared 'Fail' in the MAP examination or any other punishment which the university may decide.

Enrollment Nos.	Names	Signature
207500592036	Mahipalsinh A. Magodra	
207500592038	Avani P. Kachhadiya	
207500592046	Isha D. Gorasiya	
207500592056	Shivani B. Jasoliya	
207500592085	Saurabhkumar B. Kansagara	

Place: Surat Date:

Company / Organization Certificate

Plagiarism Report

Institute Certificate

Acknowledgement

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the Leaders, whose constant guidance and encouragement crown all the efforts with success.

We are grateful to **Gujarat Technological University** to include this Summer Multi-Disciplinary Action Project as part of curricular of MBA Program that helped us to gain practical knowledge about corporate world.

We express my thanks to **S.R. Luthra Institute of Management** and our honorable Director **Dr. Jimmy Kapadia** for providing opportunity to take Summer Internship in the industry to have practical exposure.

It is our privilege to express my deep sense of gratitude to my mentor **Dr. Roshni Singh** for his efforts, guidance, valuable comments and suggestions for making this project report.

We are thankful to **Mr. Yashpal Patel** (owner), **Mr. Mayank Patel** (Plant Head) & **Staff of Vijay Dairy** for providing us the opportunity to have the grateful experience and learning at Dream Creation.

Finally, we express our intense gratitude to our **Parents** whose blessings have helped us translate our efforts into fruitful achievement & to all of them who have directly or indirectly helped us to undergo this project.

NAME	ENROLLMENT NUMBER
Mahipalsinh A. Magodra	207500592036
Avani P. Kachhadiya	207500592038
Isha D. Gorasiya	207500592046
Shivani B. Jasoliya	207500592056
Saurabhkumar B. Kansagara	207500592085

EXECUTIVE SUMMARY

The project was taken into consideration for the promotion and the marketing of the Vijay dairy new product i.e. “Probiotic curd”.

The main task assigned to us by our mentors to make people aware about probiotic curd .It also focuses on the competitors’ product like Amul Probiotic dahi, Nutrifit probiotic yogurt etc. The main aim of the multidisciplinary action project was to first analyses the market situation focus on the need of the consumers what the thing they are lacking in the competitors’ product so that the same thing didn’t happened in our product. So, after the whole promotion and marketing of the product. And also estimate cost for the developing and launching probiotic curd.

Vijay dairy also introduce modified atmosphere packaging(MAP) to increase dairy product life to approximately twenty-five to thirty days till it packed. This study will help Vijay dairy to estimate its cost of MAP as well as they know which kind of packaging customer will accept. Vijay Dairy also faces the problem of skilled labors due to covid-19 as most of customers are from Rajasthan, Bihar, and Utter Pradesh So, because of pandemic they return to their hometown so there is shortage of labors.

Vijay Dairy major four problems we identify and give them practical solution for that. We discuss with owners of the business and give viable solutions for the problem they facing and implements as well. After implementation phase we measure the effectiveness of the solutions so take corrective decision on time and the results of the solutions are effective for the Dairy.

Table of Content

Sr. No.	Particulars		Page No.
I	**Title Page**		I
II	**Company Certificate**		II
III	**Students Declaration**		III
IV	**Plagiarism Report**		IV
V	**Institute Certificate**		V
VI	**Acknowledgement**		VI
VII	**Executive Summary**		VII
VIII	**Table of Content**		VIII
1.	**Introduction of Study**		1
2.	**Industry Profile of Food Industry**		2-20
	Introduction of a Food Industry	2	
	Global Scenario of Food Industry	4	
	National Scenario of Food Industry	9	
	State Scenario of Food Industry	14	
	PESTLE Analysis of Food Industry	18	
3.	**Profile of Vijay Dairy**		21-33
	Introduction of Vijay Dairy	21	
	Vision	22	
	Mission	22	
	About Vijay Dairy	22	
	Product Portfolio	24	
	SWOT Analysis	26	
	Introduction of Probiotic curd	28	
	Modified Atmosphere Packaging (MAP)	32	
4.	**Diagnosis phase**		34-36
5.	**Design and analysis phase**		37-38

6.	**Implementation Phases**	**39-42**
7.	**Future Recommendations**	**43**
8.	**Annexure**	**44-59**

INTRODUCTION OF STUDY

Developing and successfully launching new products is an important part of a small business growth strategy. New products provide additional sources of revenue by enabling the business to sell more to existing customers, attract new customers or enter into new markets. However, product development can be a costly, time-consuming activity. It, therefore, carries risk as well as reward. To succeed, businesses must follow a structured development and launch process.

Packaging is important in food industry as many foods spoil rapidly in air due to moisture loss or uptake, reaction with oxygen and the growth of aerobic micro-organisms i.e., bacteria and molds. Microbial growth results in changes in texture, color, flavor and nutritional value of the food. These changes can render food unpalatable and potentially unsafe for human consumption. Storage of foods in a modified gaseous atmosphere can maintain quality and extend product shelf life by slowing chemical and biochemical deteriorative reactions and by slowing or in some instances preventing the growth of spoilage organisms.

Dairy is a ubiquitous part of food life, ever present in both food service and packaged goods, whether as an end product (such as milk or yogurt) or as a critical input for iconic products such as cheese pizza. Given its intrinsic presence, dairy is a microcosm of the food industry, with the preferences of dairy consumers largely being influenced by the same trends affecting the broader food sector.

Hence the research study focuses on the title **"New Product Development and Improvement of Packaging at Vijay Dairy".** This study will help to understand the consumer preference for probiotic curd and improvement of packaging. This would further enable the Vijay dairy to strategize their business customer and estimate its cost for probiotic curd as well as packaging of dairy product as dairy product life is very short maximum two to four days but modified atmosphere packaging(MAP) increase its life to approximately twenty five to thirty days till it packed.

INTRODUCTION OF FOOD INDUSTRY

Introduction:

The food industry is a complex, global network of diverse businesses that supplies most of the food consumed by the world's population. The term food industries cover a series of industrial activities directed at the production, distribution, processing, conversion, preparation, preservation, transport, certification and packaging of foodstuffs. The food industry today has become highly diversified, with manufacturing ranging from small, traditional, family-run activities that are highly labor-intensive, to large, capital-intensive and highly mechanized industrial processes. Many food industries depend almost entirely on local agriculture, produce, or fishing.

It is challenging to find an inclusive way to cover all aspects of food production and sale. The UK Food Standards Agency describes it as "the whole food industry – from farming and food production, packaging and distribution, to retail and catering." The Economic Research Service of the USDA uses the term food system to describe the same thing, stating: "The U.S. food system is a complex network of farmers and the industries that link to them. Those links include makers of farm equipment and chemicals as well as firms that provide services to agribusinesses, such as providers of transportation and financial services. The system also includes the food marketing industries that link farms to consumers, and which include food and fiber processors, wholesalers, retailers, and foodservice establishments.

Probiotics a new entry in the market of functional foods has been defined by various organizations in various ways. According to FAO/WHO, 2002 “Probiotics are living microorganisms which when administered in adequate amounts confer health benefits to the host”. The Natural Health Products (NHP) Regulations define a probiotic as a monoculture or mixed culture of live micro-organisms that benefit the microbiota indigenous to humans. The property of probiotics showing relation between nutrition and health with negligible side effect keeps it on upper hand among all functional foods. With change in era of lifestyle people have become more health conscious thus leading

to extension of probiotics market. Attitude of consumer towards a product can be divided into three components: the cognitive component which defines the beliefs of consumer towards a product, Affective component which defines the feelings or emotions of consumer towards the product obtained from effective experience and the cognitive component defines the probability that the individual will behave in a specific way. Thus, attitude of today's consumer towards a product is affected not only by sensory properties but is overlapped by its health properties and knowledge of functional ingredients of the products.

Although status of probiotics is still not clear in health industry as they have applications both as preventive as well as curative therapy, for instance probiotics are available in various varieties in the market such as Functional food & beverages (dairy products, non-dairy beverages, breakfast cereal, baked foods, fermented meat products, dry-food probiotics), dietary supplements (food supplements and nutritional supplements), Specialty nutrients (infant formulation), pharmaceutical preparations and as Animal feed.

Thus, to create a stronger market of probiotics it is required to strengthen the positive attitude of consumers towards the products containing probiotics, which can only be attained if the people understands the benefits of the product on their health. Investigation of knowledge, thoughts, beliefs, behavior and feelings of people can help to determine the current status of products containing probiotics and the pace of market development of probiotics products in a certain region.

Global Scenario of Food Industry:

The food industry is a complex, global network of diverse businesses that supplies most of the food consumed by the world's population. The term food industries cover a series of industrial activities directed at the production, distribution, processing, conversion, preparation, preservation, transport, certification and packaging of foodstuffs. The food industry today has become highly diversified, with manufacturing ranging from small, traditional, family-run activities that are highly labor-intensive, to large, capital-intensive and highly mechanized industrial processes. Many food industries depend almost entirely on local agriculture, produce, or fishing.

At present, the global market is partially challenged by the logistical and associated economic impacts of the pandemic. Regardless of these external factors, the demand for health and immune-boosting products is predicted to grow at a healthier pace. Currently, buyers are more aware of dietary supplements and functional foods, and the important role these products play in ensuring better health and wellbeing. Therefore, these products are becoming more popular owing to their various health benefits such as enhancement of gut health and improvement of immune response, which will aid the market growth in the near future.

Several large, as well as small manufacturers and retailers have reported a surge in the demand for these products in recent months. For instance, in April 2020, Probi, a global leader in probiotics, reported a considerable surge in the sales of its products that was clearly a result of the increased interest in immunity products from the consumers. This demand is anticipated to remain incredibly high during the next few months as individuals have become gradually conscious of their health and well-being owing to the sudden outbreak of coronavirus.

Latest Trends

Escalating Adoption of Probiotics as a Replacement for Pharmacological Solutions to Surge Demand

Probiotic bacteria fortified food has become popular among buyers who are seeking natural solutions to treat numerous health ailments. These products can act as a substitute for various pharmaceutical solutions as they are considered as an effective natural way in treating gastrointestinal conditions and gut health. The increasing evidence regarding the health benefits linked with the product's consumption and the growing popularity of probiotic strains as a natural, cost-effective, and safe substitute for pharmacological solutions are projected to positively impact the market growth.

Driving Factors

- **Rising Adoption of Immunity Boosting Probiotics Fortified Food to Favor Growth**

  Probiotics help in the digestion process along with numerous health benefits such as boosting immunity, maintaining gut health, weight management, preventing obesity, and many others. The increasing awareness about the health benefits offered is anticipated to drive the market in the forthcoming years. The rising health-consciousness among consumers and the emerging trend of veganism are luring consumers towards a plant-based diet which is expected to boost the demand for plant-based products. As a result, several niches as well as prominent market players are embarking on the launch of plant-based

probiotics to increase their consumer base. For instance, in August 2020, Chobani, an American Food Company announced the launch of a new probiotic drink. The new product is available in four different flavors i.e. Lemon Ginger, Pineapple Turmeric, Peach Mint, and Cherry Hibiscus Tea.

- **Increasing Investment in R&D Activities to Promote Growth**

  The increasing investment by prominent players on the research and development of innovative products is expected to positively impact the probiotics industry growth. For instance, in February 2020, Amor pacific Group opened its novel Green Tea Probiotics Research Center to expand the study of lactobacillus (a beneficial microbial strain) recently found in Jeju organic green tea garden.

  The companies are actively focusing on mergers and acquisitions to diversify their product portfolio and consumer offerings. For instance, in April 2020, Chr Hansen announced the acquisition of HSO Health Care. This acquisition of the Austria-based company specializing in probiotics for women's health can help to strengthen Chr Hansen's probiotic offerings.

Restraining Factors

- **Risk of Contamination to Hamper the Growth of the Market**

  Probiotics are the live microorganisms and the probiotic-containing products are wholly dependent on the cultivation and secretion by probiotic living organisms. The risk of contamination of these products by the non-desirable microbes with high toxicity is expected to hamper the production of a specific batch. Moreover, the high investment required for the research and development of new bacterial strain is predicted to further limit the market growth.

Segmentation

By Microbial Genus

Lactobacillus to Hold a Major Share of the Market Owing to their Diversified Applications & Benefits

Lactobacillus held the largest share in the global probiotics market in 2019. Lactobacillus strains have an inhibitory potential to minimize the growth of microbes and pathogens, which is favorable for extending shelf-life and increasing the hygienic superiority of foods. Lactobacillus strain is beneficial to humans because of their potential in reducing enteric diseases in humans, which is projected to positively impact product demand.

Yeast is expected to grow at the highest CAGR in the global market. Saccharomyces boulardii is one of the widely utilized yeast for the production of probiotic drinks. It helps to regulate the intestines and protect them from harmful pathogens that can damage the intestinal lining. It further regulates several parts of the immune system and keeps the intestinal barrier function up and running. The rising awareness regarding the benefits associated with the consumption of these products is expected to positively leverage product sales.

By Application

Functional Food & Beverage Segment to Dominate the Global Marketplace Owing to Increasing Demand

At present functional foods and beverages have become a part of an everyday diet and are demonstrated to offer potential health benefits beyond the widely accepted nutritional effects. Currently, the consumption of functional food and beverages is one of the trending foods habits worldwide. The increasing popularity of natural and health-benefiting food products is driving product sales. The rising awareness among consumers regarding the good and balanced diet benefits in controlling disease risk and maintaining overall health is expected to boost the demand for dietary supplements that meet their needs.

Regional Insights

Increasing Demand for Fermented Drinks in Asia Pacific to Favor Growth

Asia Pacific is the largest market at present and is expected to account for a significant

proportion during the forecast period. The region has evolved rapidly over the last decade with China and Japan occupying more than half of the regional sales. The demand for probiotic products, especially drinks in Japan and some countries in Central and Southeast Asia is expected to remain promising during the forthcoming years. For instance, Japan-based Yakult Honsha, claims that the company sells 9,540 Yakult bottles per day in Japan. The increasing popularity of these fermented drinks is expected to support regional market growth.

North America is predicted to witness considerable growth in the global market. The well-established food industry, presence of prominent players, and rising prevalence of lifestyle-related diseases such as diabetes and obesity are encouraging the consumers to inculcate healthy eating habits, which in turn, is likely to boost the product sales in the region.

Key Industry Players

Industry Emphasis on New Product Development to Aid Market Growth.

The growing consumer's understanding of these products and its effects that go beyond digestive health is likely to increase the product demand at a healthier pace. The thriving demand for this product is motivating market players to diversify its product portfolio and distribution network to increase their market reach and consumer base.

Danone S.A. is one of the leaders operating in the market. Innovations and expansion of business are the key strategies adopted by the company to compete in the global market. Danone announced the launch of new plant-based variants for its Danio and Activia yogurt brands in the Netherland market. This launch is a completely new experience for Danone as the company for the very first time is concentrating on products that are 100% plant-based. Danone North America announced the launch of its new kids' yogurt brand Danimals. The newly launched low-fat yogurt contains vitamins C and D, along with probiotic products that claim to potentially aid the immune system of kids.

National Scenario of Food Industry:

India is the second biggest food producer globally, and it has the potential to become the largest producer of food in the whole world. The Food Industry in India includes milk and milk products, plantation, alcoholic beverages, vegetables and fruits, fisheries, poultry and meat, grain processing, chocolates and confectionery.

The Food Industry in India exported products worth around US$ 5.8 billion in 1998, while the total world food export stood at US$ 438 billion. This shows that although India is one of the biggest food producers globally, it accounted for only 1.5% of the international trade in food as in 1998.

Through the Ministry of Food Processing Industries (MoFPI), the government endeavors to urge more investments in the business. It has sanctioned joint ventures (JV) proposals, foreign collaborations, industrial licenses and 100% export-oriented units. The Indian food processing industry amounts to 32% of the country's aggregate food market. It contributes approximately 8.80 and 8.39% of Gross Value Added (GVA) in Manufacturing and Agriculture, respectively, is 13% of India's exports and 6% of aggregate industrial investment.

Reasons for the growth of the food industry

- Increased urbanization in the country has given a boost to the food industry in India. The majority of the women in the country are working because families have extra income, which they are spending on buying food.
- The standard of living in the country has improved as a result of which there has been an increased demand for food.
- Supermarkets and malls have opened all over the country, and this has helped in making shopping for food is a pleasurable experience.
- The online food delivery players such as Food Panda, Zomato, TinyOwl and Swiggy build scale through partnerships as the organised food business has a vast potential and a promising future. In 2016, the online food delivery industry thrived at 150% year-on-year with an estimated Gross Merchandise Value (GMV) of US$ 300 million.

- The consumers can select, inspect and pick up food items that they like in a comfortable ambience. This, too, has helped to boost the sale of food products in the country.

Government Initiatives

Some of the major initiatives taken by the Government of India to improve the food processing sector in India are as follows:

- The Government of India aims to boost growth in the food processing sector by leveraging reforms such as 100 per cent Foreign direct investment (FDI) in marketing of food products and various incentives at central and state government level along with a strong focus on supply chain infrastructure.
- In Union Budget 2017-18, the Government of India has set up a dairy processing infra fund worth Rs 8,000 crore (US$ 1.2 billion).
- The Government of India has relaxed foreign direct investment (FDI) norms for the sector, allowing up to 100 per cent FDI in food product e-commerce through automatic route.
- The Food Safety and Standards Authority of India (FSSAI) plans to invest around Rs 482 crore (US$ 72.3 million) to strengthen the food testing infrastructure in India, by upgrading 59 existing food testing laboratories and setting up 62 new mobile testing labs across the country.
- The Indian Council for Fertilizer and Nutrient Research (ICFNR) will adopt international best practices for research in fertiliser sector, which will enable farmers to get good quality fertilisers at affordable rates and thereby achieve food security for the common man.
- The Ministry of Food Processing Industries announced a scheme for Human Resource Development (HRD) in the food processing sector. The HRD scheme is being implemented through State Governments under the National Mission on Food Processing. The scheme has the following four components:

  - ✓ Creation of infrastructure facilities for degree/diploma courses in food processing sector

- ✓ Entrepreneurship Development Programme (EDP)
- ✓ Food Processing Training Centers (FPTC)
- ✓ Training at recognised institutions at State/National level

Probiotic Food:

Probiotics are functional foods that go beyond providing nutrition to the human body. While the use of probiotic drugs is widely accepted in India, intake of probiotics as a food supplement is yet to gain significance. People today are more informed and prefer to take the preventive path rather than adopting a curative approach. Given this approach, the future of the probiotic industry in India, which is still at a very nascent stage, looks promising.

What are probiotics?

Probiotics are live microorganisms, which, if consumed in ample quantity, are said to be favorable to the host. Probiotics are mostly present in the form of bacteria, moulds and yeast, with bacteria being the most widely used. Probiotics can be consumed in the liquid state, as a powder, a gel, a paste, granules, capsules, sachets, etc. Live microorganisms improve intestinal microbial balance which spells better health and immunity.

Global essence, India story

The global market for probiotic foods and drinks is pegged at $32.6 billion, with Europe and Asia contributing to 42 per cent and 30 per cent of the total revenues. The industry estimates suggest the growth of the industry at a CAGR of 12.6 per cent, with the increase said to be for a span of four years beginning 2009.

Introduced in India in 2007, its current stature stands at $10 million with growth figures indicating an 8 to 10 per cent increase for the past 3-4 years. Munish Soni, Deputy General Manager, Marketing, Dairy Products Division, Mother Dairy, says, “The Indian probiotic industry is miniscule and at a very burgeoning stage. Currently, it stands at about Rs 50 crore and is growing at 20 per cent.” The Indian contribution to the world market for probiotics, however, is less than one per cent in terms of turnover.

Brand value

Probiotics are found mainly in milk and fermented milk products. This industry is stated for immense growth, considering the fact that India is the largest producer of milk across the globe and has the world's highest cattle production. However, the potential of the probiotic's category is yet to be realised fully. There are just a handful of brands operating in the segment and the product range is limited to curd, ice creams and milk-based drinks. The major players in the category are Nestle, Amul, Mother Dairy and Yakult.

Amul introduced Prolife Probiotic Ice Cream and Sugar Free Prolife Probiotic Ice Cream in 2007, followed by Prolife Lassi in sachets and then in plastic cups. In 2011, it introduced Flaavyo Fruit Yoghurt, and this year, it has launched Flaavyo Frozen Yoghurt. The most important facet of Amul probiotic products is that the brand uses world class ingredients in terms of probiotic culture, milk, fruit/pulps, etc.

Sold globally as a fermented milk drink, Yakult contains more than 6.5 billion beneficial bacteria (Lactobacillus casei strain Shirota) in a 65 ml bottle. Priced at Rs 10 for a bottle, these bacteria reach our intestine alive and impart various health benefits.

Mother Dairy offers Pro Active Dahi and a drink. The price range commences at Rs 10 and their products also contains dietary fiber.

Retail outlook

The sale of probiotic products is limited to metros and some of the Tier I cities only. Rural areas are not on the list as the product calls for some concerns, which cannot be met at that level, the basic being refrigeration. Some areas do not even have access to electricity so it becomes impossible to offer shelf space to these products.

"Yakult is available in all metros except Kolkata. We have made the product available in south India and in the next phase, we are targeting Tier II cities. As the category grows, our reach will also increase," Sapra says. Yakult, which is bought 2-3 packs at

a time, has the strongest sales in Delhi-NCR, taking into account that it is their oldest market

Maintenance concerns

Probiotic products have a specific shelf life and need to reach the retailer and customer to be suitable for consumption. This calls for apt distribution and cold chain facilities for their transportation, which is a concern for this industry. Yakult has in-house cold chain management solutions and has its own trucks that transport the product at controlled temperatures of 10 degrees Celsius to retail locations. The Yakult ladies use red chiller boxes, which contain chiller packs that help control the product temperature.

Marketing initiatives

Mother Dairy is targeting kids for its probiotic products, as these products boost immunity. For the same, it has partnered with schools and introduced awards for those who have the best attendance. Also, it has on board dieticians, who prepare charts and activities for the kids. Mother Dairy also has a doctor engagement strategy that it follows to connect with the customers indirectly. Of the total curd business, probiotic curd contributes to about 7- 8 per cent.

Yakult realised the potential of reaching out to the customer in the right way and has worked towards it. Kiyoshi Oike, Managing Director, Yakult Danone India, says, “We are trying to build awareness about this category through a TV commercial that has Bollywood actor, Kajol, as our brand ambassador and other BTL activities. We also do direct sampling with consumers at their homes, where our Yakult ladies (we also call them ambassadors of good health) explain the benefits of probiotics to consumers.”

Amul is targeting youth and health conscious people and has promoted its probiotic range through press, hoardings and various BTL activities. To overcome the awareness hurdle in this industry, it has converted its chocobar into prolife ice cream. Sodhi says, “Anyone who wants to consume Amul Chocobar will consume probiotics and this is helping us to spread awareness about the benefits of probiotics.

Future intensification

The road ahead seems great for this industry. All the aspects that are required for the augmentation of this industry are well present in India and therefore, its escalation is inevitable. Stating figures, Soni says, "The industry is growing at 25 per cent every year and in the future, the industry will grow by leaps and bounds. As long as people prefer foods with benefits, the industry is poised for growth." Oike adds, "The global probiotics industry is estimated to reach $19.6 billion by 2013."

State Scenario of Food Industry:

Gujarat, one of the most developed states of India, offers diverse set of investment opportunities in food processing sector. A bird's eye view of potential project for investments opportunities in food processing sector in Gujarat is presented here.

Gujarat, located on the Indian Peninsula's western coast, is the nation's most industrialized state. It shares borders with Rajasthan and Madhya Pradesh, Maharashtra, and the Union Territories Daman and Diu, Dadra and Nagar Haveli. The Arabian Sea borders the state to both the west- and south-west. The state's coastline is 1600 km long and contains many natural harbors. It also hosts large ports such as Kandla or Mundra, which are the largest private ports in India, it is well connected with the rest of India.

Why Gujarat

- Gujarat accounts for the largest share in the total investments in the food processing sector of India.
- Gujarat is keen to promote the agro processing industry.
- Gujarat has 51 dairy plants, the second highest in the country, besides known for its brand 'Amul.'
- Gujarat's 1600 km long coastline, which is broken by several bays, inlets, estuaries and marshy lands, is available for fishing activities.

Raw material Base

Gujarat is one of the most developed states of India. Its agricultural strength lies in its

diverse climate, robust marketing infrastructure and enterprising farmer base. More than half of the state's land is under cultivation. Gujarat's agricultural economy is shifting towards high-value crops such as cotton, groundnuts, fruits, vegetables, condiments, and spices. Gujarat is the largest producer of groundnuts in the country and the third-largest producer of milk.

Food Grains production

Total production of 62.33 lakh MT of food grains

- Wheat - 2.4 million tonnes
- Pearl millet - 790 thousand tonnes
- Pigeon pea (Tur Dal)- 237 thousand tonnes

Horticulture products production

Total production of 216 lakh MT of horticulture products

- Fruits 8.2 million tonnes
- Vegetables - 12.6 million tonnes
- Spices 0.72 million tonnes

Livestock production

- Total production of 122 Lakh MT of milk, ranked second in milk production
- 8.82 million milch animals in the state as per 19th livestock census
- Gujarat's contribution in the National Marine Fish production is 20% in the year 2014-15 and its share in the GDP of the State is around 0.84 %

Agriculture marketing

Gujarat has the following marketing facilities in the state:

- Food Processing Units: More than 30,000 units in Food Processing Sector in Gujarat are operating providing employment to over 1 million people.
- Agri-Export Zones: Total of 20 Agri-export zones, with focus on mango and vegetables and sesame seeds have been identified by the State Government, 15 for Mango and Vegetables and five for sesame seeds o There are 194

Agriculture Produce Market Committee (APMC) market yards and sub market yards in the state.

- 560 cold storage and fish processing units
- with capacity of 20,30,873 MT
- 207 warehouses across 45 centers
- One irradiation unit in Bavla, Gujarat
- One Air cargo complex for Food exports
- Delhi Mumbai Industrial Corridor which passes over Gujarat

Dairy Sector in Gujarat

- Gujarat is one of the largest milks producing states in India with the contribution of 7.75% share in the total milk production of India.
- The state having 17 Cooperative dairy milk unions & 25 private dairy plants has a milk collection of 3.45 billion liters with over 3million milk producers, affiliated to more than 15,000 Primary Milk Cooperative Societies.
- Milk contributes to 22% to the Agricultural GDP of Gujarat and is one of the biggest sectors for supporting livelihood in the state.

There are more than 10,000 companies in Food Processing Sector in Gujarat. An indicative of key companies and their district of presence in the state is given below.

State Government Initiatives

The vision of Gujarat's comprehensive Agro Business Policy is "To ensure food and nutritional security; and economic empowerment to the people through accelerated yet sustainable growth in agriculture". The main objectives of the Agri business Policy 2016-2021 are:

- To increase investment opportunities in food processing and agricultural industries
- To increase agricultural production and productivity
- To increase farmers' income
- To create more employment opportunities in rural areas through agribusiness
- To provide the best training for skill development in the field of agribusiness

- To increase processing and reduce wastage
- To increase their exports by maintaining high standards of agricultural produce
- To give highest priority to food security and nutritional security
- To promote agricultural practices that preserve the environment.
- To give advance opportunity to youth and women in agro business.

Organic Milk, Milk Product Manufacturing and Processing Unit of Gujarat:

A unit for the production of organic milk and other milk products in Gujarat is proposed. This project will integrate modern animal husbandry and a modern milk processing plant to produce a range of organic products. It is designed to ensure that the milk remains uncontaminated by air and human hands. The whole process will be automated in a sealed vacuum environment.

PESTAL ANALYSIS OF THE FOOD INDUSTRY

Political:

Here are the Political factors impacting the food industry:

- ✓ **Wide Regulation**

  Governments across the world have expansive regulatory frameworks for every aspect of the food industry. This includes the cleanliness of commercial kitchens, the standards for storing and transporting produce, and even the requirements for laborers in the food business. Without a doubt, this makes the food industry one of the most tightly-regulated industries of all. On the plus side, this ensures that consumers aren't exposed to poor quality nutrition, but the complexities of regulation certainly take away from the margins of the food business.

Economic:

Here are the Economic factors impacting the food industry:

- ✓ **Growing Disposable Incomes**

  As a general trend, the world's population is only getting richer. That means that individuals in the lower, middle, and upper classes all have more money to spend on luxuries — including restaurant food. As a result, the overall revenue of the food industry is growing, as individuals cook less and eat out more often. This has a positive effect on all corners of the space, including restaurateurs, food distributors, and the individual workers who play a role in these businesses.

- ✓ **Increasing Labor Costs**

  Disposable incomes are growing for a reason: laborers are earning more money these days. On the whole, the cost of hiring workers is increasing across all industries. This is caused by not only a growing demand for employees, but also higher and higher government expectations for minimum wages. As in many other industries, the effect of increasing labor costs is simple: less margin for the owner of the business, and thus less profit.

Sociocultural:

Here are the Sociocultural factors impacting the food industry:

- ✓ **Health Consciousness**

  Nowadays, scientists know more about the relationship between food and our bodies than ever before. There's a clear relationship between the food we eat and our personal health, and consumers are conscious of this. As a result, many individuals are looking for healthier ways to fuel their bodies. This doesn't necessarily have a positive or negative effect on the food industry, but it means that businesses will have to adapt to stay relevant. For example, fast food businesses will likely have to move away from traditional, high-calorie fried foods towards healthier alternatives like salads.

- ✓ **Dietary Restrictions**

  Aside from having a better grasp of what kinds of food are and aren't healthy, consumers are also more knowledgeable about their individual dietary restrictions. For example, many individuals now understand the negative impact of gluten in those with Celiac disease. This has led to consumers expecting greater understanding on behalf of those who work in the food industry. Once again, this isn't necessarily a bad thing, but it means that the food industry will have to make changes to keep clients happy.

Technological:

Here are the Technological factors impacting the food industry:

- ✓ **Automation**

  We're seeing various types of automation more and more in the food industry. Perhaps the best example is the use of self-checkout screens at fast food venues such as McDonalds, but it's not the only one! Just recently, social media platforms went crazy as viral footage of a hotel's robot cooking up omelets began to spread. As we find more ways to use technology — including robots — in the food industry, there will be less need for laborers. Overall, this is a good thing for the industry, as it will allow businesses to improve profitability and reduce the likelihood of human error.

Legal:

Here are the Legal factors impacting the food industry:

- ✓ **Safety Standards**

  As touched upon in the Political section of this PESTLE analysis, the food industry has high standards for safety matters. In particular, there are scores of rules in every country on how food should be transported, stored, and prepared including directions on what temperatures various food types can reach, how they should be cleaned, and so on. While this is indeed largely a Political issue, it becomes a Legal matter if any of these regulations are ever breached.

Environmental:

Here are the Environmental factors impacting the food industry:

- ✓ **Impact of Meat**

  Not only is there growing awareness for the health repercussions of the food we eat, but also for the environmental repercussions of the food we eat. One particularly problematic food group from an environmental point of view is meat. The production of meat — especially red meats — uses huge amounts of water and creates a significant carbon footprint. No less, the meat industry is tearing down large amounts of forest to create new space for farms. The result of this is that more and more individuals are switching to plant-based diets, and governments are slowly taking interest. Once again, this isn't necessarily a negative for food businesses, but they will have to recognize the impact of this shift in the long term.

Conclusion:

This PESTLE analysis of the food industry is certainly an interesting one. It's a mix of positives, negatives, and uncertainties. On the one hand, consumers have more to spend on food and robots can reduce expenses. On the other hand, the space is carefully regulated and labor prices are increasing. What's more, eaters' diets are becoming increasingly more specific. Of course, the food industry is here to stay, but it seems those who prevail will have to understand what consumers really want to be eating in the 21st century.

PROFILE OF VIJAY DAIRY

INTRODUCTION OF VIJAY DAIRY

Name of Dairy	Vijay Dairy
Established	1975
Cuisines	Desserts, Street Food
Type	Quick Bite Outlet, Sweet Shop
Founder	Late Shri Jagjivandas Govindji Patel
Owner	Mr. Yashpal Patel
Plant Head	Mr. Mayank Patel
Total Branches	Company outlets- 5 Franchise- 9
Plant Location	Vijay Dairy Products Plot No. A/38, Ichchhapor GIDC, Hazira - Magdalla Road, Surat - 394 510 www.vijaydairy.com
Main Branch	U/4 Ramsha tower adajan patiya surat-395006 Contact no.: 8980955888 www.vijaydairy.com

VISION

"We are socially responsible organization with commitment to consumer delight and respect for all. We believe in growth and continuous improvement through teamwork, trust & excellence without compromising our honesty and integrity"

MISSION

- To provide maximum return to producers
- To Provide good quality Milk and Milk Product to consumers
- To increase milk productivity at most economic rate by providing technical input services to the producers.
- Factory expansion.

ABOUT VIJAY DAIRY:

"VIJAY DAIRY" is brainchild of Late Shri Jagjivandas Govindji Patel, father of four brothers who later on managed operations of the group. The "VIJAY DAIRY" was started in 1975

Although the beginning was rough for this family owned and operated business, Vijay Dairy has gained popularity by word-of-mouth publicity. From beginning, promoters were very clear in mind that they want to provide the best and quality food products to the society. Thus, giving their contribution in creating healthy society.

Vijay Dairy has its own milk collection centers. This center collects raw milk from milk producers directly. Company has placed BMC (Bulk Milk Cooler) units at distant location to ensure quality of milk. Transport vehicle collects this milk and bring it to processing unit. At processing unit, milk instantly pasteurized and diverted to different department for manufacturing other milk products.

New Product Development of Probiotic Curd and Improvement of Packaging at Vijay Dairy

Today, Vijay Dairy is a leading manufacturer and distributor of Milk, Milk Products like Paneer, Ghee, Lassi, Matho, Khoa, Khoa Sweets, Dry fruit Sweets, Bengali Sweets, Besan Sweets, Namkeens and Snack items.

Vijay Dairy has 5 retail outlets in Adajan - Pal area of Surat city, 1 milk collection center and 1 manufacturing unit. Vijay Dairy's Group turnover is 51 crores (approx.). We are also getting franchisee opportunities. These days, we are expanding our manufacturing unit to Ichhapore GIDC.

Now our NRI customer base is also increasing and we give them milk and milk products in best packaging, so that the products remain good even after long time.

PRODUCT PORTFOLIO:

Milk, Butter milk, Curd, Ghee, Butter:

Paneer, Lassi, Shakes:

Sweets:

Fast Food:

SWOT ANALYSIS OF VIJAY DAIRY:

Strengths:

- Largest pool of dairy cattle in the world.
- Availability of skill for animal rearing in most parts of the country.
- Indian breed of cows represented the world's largest milk producing herd.
- India is largest producer of milk in the world.
- 70% of milk production is done by small & marginal farmers in the country & is larger employment provider in the non-farm sector.

Weakness:

- Coexistence of large portion of useless cattle population with low productivity.
- Too much fragmented production of milk.
- Labour is becoming costlier.
- Traces of insecticides, pesticides &antibiotics are common in milk.
- Poor knowledge about nutrition & balanced ration for animal.
- Poor personal hygiene of farmers & his family & ignorance of clean milk production also leads to poor quality of milk.
- Poor infrastructure for milk collection, chilling & transportation.
- Poor infrastructure in networking of information.

Opportunities:

- Huge demand of milk & milk product in unorganized retail.
- High purchasing power of customer.
- Large corporate is looking for developing larder farms. More awareness about benefits of cow's milk.
- Large opportunities in carbon credits & development of nonconventional environment friendly sources of energy through dairying.

Threats:

- High rate of urbanization.
- No distribution & pricing policy on milk.
- No control on unproductive animal's growth both for bulls & cows.
- No incentive on clean milk production.

Introduction of Probiotic Dahi:

A probiotic is a food substance that contains and promotes the growth of beneficial bacteria in your body2. Some of the probiotic foods are yoghurt, curd, fermented vegetables (sauerkraut- like soaked in vinegar, pickled), fermented cheese (cheddar and swiss) and sour pickles.

The curd is a milk product that contains the sugar lactose and protein casein. The Lactobacillus bacteria grow on lactose and produce lactic acid as a byproduct of fermentation. The lactic acid produced decreases the pH of the milk which curdles the milk protein casein.

Homemade curd contains several species of the Lactobacillus like L. fermentum, L. acidophilus, L. confuses, and Leconostoc lactis is also present in curd prepared at home.

Curd chiefly contains the lactic acid bacteria genus that converts lactose into lactic acid. A probiotic on the other hand is loaded with a wide variety of beneficial bacteria. Though curd is not truly probiotic, its benefits are very similar to that of probiotics.

What Is Gut-Friendly Bacteria?

Gut-friendly bacteria is what makes curd probiotic. Unlike the bacteria that does no good, gut-friendly bacteria helps in better digestion. While regular curd like Heritage Total Curd has all the health benefits, you should include Heritage's Probiotic Curd in your everyday diet as it contains more gut-friendly bacteria.

Health Benefits of Probiotic Dahi

- ✓ **For Stronger Bones & Teeth**

  Consuming probiotic curd daily can give us a lot of calcium and vitamin D which is essential for bone health. Just like milk, a cup of curd can do good for your kids. Do you know someone who has joint pains? Having curd every day can reduce the risk of arthritis, osteoporosis, and rheumatoid arthritis by providing enough calcium to the body.

- ✓ **For Weight Loss**

  It's no secret that a protein-rich diet helps weight loss, but did you know you can achieve that by eating curd? Curd is a great source of protein, and the gut-friendly bacteria in probiotic curd helps control weight gain. While the protein will keep your stomach filled for a long time, the calcium, along with probiotic bacteria, will stop fat cells from forming. Isn't that amazing?

- ✓ **For Anti-Aging Purposes**

  A common reason for faster ageing is nutritional deficiencies. A cup of probiotic curd everyday can give you ample amounts of calcium, zinc, B vitamins, and protein which keeps the skin firm and supple. Remember seeing DIY curd face masks? Don't just mask it, eat it!

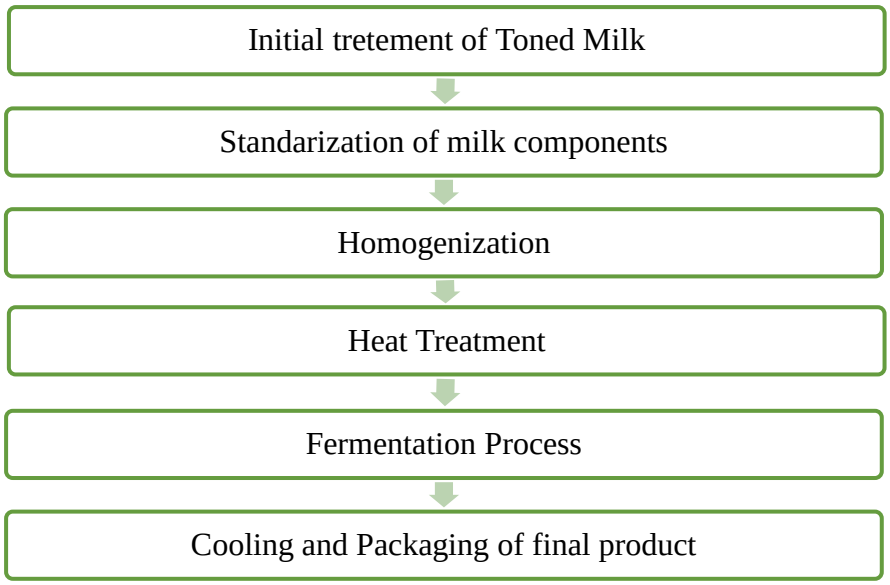
- ✓ **For Stronger Immunity**

  With a cup full of probiotic curd every day, you are ready to take on the world. Not only will the probiotic strain in curd strengthen your gut health but also reduce the chances of infections and inflammatory bowel diseases.

- ✓ **Improve Brain Function**

  Regular consumption of dairy affects human mood and cognitive function. Having a dollop of probiotic curd keeps the brain active, helps prevent nervous breakdowns, and keeps the stress away.

Process of Probiotic Dahi:



Initial treatment of milk (optional step)

Centrifugation to remove somatic cells and solid impunities, thermalization at 60 to 69°C for 20 to 30 seconds to kill vegetative microorganisms and partial inactivation of enzymes by cooling at <5°C.

Standardization of milk components

Adjusting fat and solid-non-fat contents.

Homogenization

Milk fat globule undergoes homogenization by the application of pressure, high velocity flow of the milk or high frequency vibrations.

Heat treatment

This step is carried out by pasteurization, sterilization or ultra-heat treatment to ensure the safety of the product by reducing the number of pathogenic microorganisms to safe limits for consumption.

Fermentation process

Inoculation of milk with Dahi starter cultures (1.e., Lactobacillus acidophilus, Lactobacillus rhamnosus, Bifidobacterium animalism, Lactobacillus paracasei) and the

fermentation is conducted at temperature between 40 to 44°C and stopped when the pH reached 4.7 to 4.3.

Cooling and packaging of final Dahi product

After the pH of Dahi reaches 4.7-4.3 the Dahi is cooled to -5°C.

Modified Atmosphere Packaging (MAP)

What is Modified Atmosphere Packaging (MAP)?

M.A.P. stands for Modified Atmosphere Packaging, and is also known as gas flushing, protective atmosphere packaging or modified oxygen packaging.

MAP is a technology that was developed to maintain the freshness and attractiveness of packaged products for the duration of their projected shelf life by modifying the gas surrounding the packaged good. Modifying the ambient atmosphere of packaged foods can improve visual, textural and nutritional appeal, and is considered minimal processing compared to the application of chemicals, preservatives or stabilizers to a food product. MAP is also used to preserve the quality of pharmaceutical and medical device products.

MAP is typically created by blending gases of oxygen, carbon dioxide and nitrogen together, and then packing the food/healthcare product into a hermetically sealed package surrounded by this gas mixture. Controlling the composition of this gaseous mixture helps to meet the specific respiration needs for each packaged product, minimizes on product degradation and helps to preserve and extend the packaged product's "best if used by" date.

Today MAP technology is used largely for packaged foods like cheese, cured meats, breads and pastas, and healthcare products like surgical implants and pharmaceutical blister packs.

Advantages of MAP

- Fresh appearance
- Potential shelf life increases by 50-400%
- Product can be distributed to long distances
- High quality product
- User friendly

Disadvantages of MAP

- Added cost
- Temperature control is necessary during storage.
- Special equipment is required

Dairy Products Packed by MAP

MAP has the potential to increase the shelf life of a number of dairy products like fat-filled milk powders, fat spreads and cheeses.

Whole milk powder is particularly susceptible to the development of off-flavors due to fat oxidation. Commercially, the air in the pack is removed under vacuum and replaced with 100% N2 or N2/CO2 mix and the powder is hermetically sealed in metal cans where in product will have low levels of residual O2 (<1%). Use of O2 scavengers may also be used.

Cheddar cheese is traditionally vacuum packed while MAP is being used with high concentration of CO2 gas mixes and it is important to maintain the balance of correct CO2 level in the gas mix so as to avoid excessive pressure being put on the pack seal. The cottage cheese is a high-moisture, low-fat product that is susceptible to a number of spoilage organisms including Pseudomonas spp. Use of N2/CO2 atmospheres showed significant extension of the shelf life of cottage cheese. Gas mixtures containing CO2:N2 in the proportion of 40:60 can increase the shelf life of cottage cheese significantly.

DIAGNOSIS PHASE

Problem-1: People are not aware about Probiotic Dahi

The discussion with the owner of Vijay Dairy we get to know that Vijay Dairy wants to launch Probiotic Curd. But people are not much aware about Probiotic products. Awareness of product is very important as people get knowledge about particular product of firm. So, we can say that one of the problems faced by the firm to make people aware about Probiotic dahi & its benefits to customers.

Problem-2: People are not aware about Modified atmosphere packaging (MAP)

The discussion with the owner of Vijay Dairy we get to know that Vijay Dairy wants to introduce modified atmosphere packaging(MAP). As we know shelf life of probiotic curd is generally 12 to 14 days but MAP increases its potential shelf life by 50-400%. But people are not much aware about this kind of packaging. People have image in their mind that milk product are consumable for 12 to 14 days after packaging to break this image in consumer mind is important to make them aware about MAP that MAP packaging increases shelf life by 28 days. So, we can say that one of the problems faced by the firm to make people aware about modified atmosphere packaging.

Problem-3: Distribution of cost for Probiotic Dahi:

To survive in a today's competitive environment especially when there is a high competition in food industry. However, Vijay Dairy has a many client in surat but still it has to work hard to survive. To meet their strong clients', they need to reduce their cost to survive in market. There is already some probiotic curd in market which has pricing between Rs.24 to Rs.40. to survive in market Vijay dairy needs to manage there cost according to market.

Problem-4: Shortage of Skilled labour:

We all know that COVID-19 epidemic shutdowns, layoffs, and firm exits create many issues, one of them issue is shortage of labour. The Vijay Dairy faces labour shortage due to COVID-19. . Basically, the labour works in Vijay Dairy were from Rajasthan Bihar and Uttar Pradesh. During the lock down they went to their village. Because of 3-4 month of lockdown period they have no income. And now they have not enough money to come back again and survive in Surat. Because of this reason Vijay Dairy faces issue of labour shortage.

MAJOR OBJECTIVES AND TASK, PROJECT SCHEDULE

By analysing the above problem, we found following area where company needed to work upon, we need to focus on make aware about probiotic curd and modified atmosphere packaging. And also needs to focus on distribution of cost for probiotic curd and mange shortage of labour.

The objectives of the project are as following:

- ✓ To know people are aware about Probiotic Dahi.
- ✓ To know what Influence peoples to buy Probiotic Dahi.
- ✓ To aware people about Modified Atmosphere packaging.
- ✓ To estimate cost of probiotic curd.
- ✓ To find out the reason behind shortage of workers due to COVID-19 pandemic.

Roadmap and schedule of the task

To achieve the above-mentioned objective, we need to carry out different activities. Following are the activities which will be carried out during the project.

November	December	January
• Company visiting was done. Company overview was giving by the owner and also, he explained us the whole procedure f how he finished product procedure is being carried out. • By visiting the company on first day we observed that the layout of the company was not organized properly i.e. machines and equipment • While interacting with Mr. Mayank the plant head of Vijay Dairy. He gave us the major issue related to shortage of workers.	• Interacting with worker. • Observing the tasks performed by the workers. • Information was collected regarding the probiotic curd and modified atmosphere packaging	• We make poster for probiotic dahi and modified atmosphere packaging. • We visit branches and organise awareness campaign for Probiotic Dahi and MAP. • We make estimated cost sheet for probiotic dahi • We suggested the owner to assist the workers to give references. And those who give reference will be given bonus and also, we suggested to give place so they can rest.

DESIGN AND ANALYSIS PHASE

Problem-1: People are not aware about Probiotic Dahi

How to aware a people about probiotic products specially Dahi

- We first thought about doing a campaign by going door to door in the central area of surat. Still, due to the Covid situation, the company owner suggests researching and being aware of people who come in-store and do online research.
- We create a questioner about awareness of probiotic Dahi and collect responses from both online and offline.
- For offline, we go to the retail store and are firstly aware of probiotic Dahi and their advantage then give them a questioner.
- We also made pamphlets to make customer aware about a Vijay dairy probiotic dahi and made a strategy how to distribute a pamphlet. For that we go to societies and stores and distribute pamphlets.

Problem-2: People are not aware about Modified atmosphere packaging (MAP)

How to aware people about Vijay dairy's Modified Atmospheric Packaging

- We also identify that the Self-life of milk-related products is 12 to 14 days only. For this problem, Vijay dairy uses Modified Atmospheric Packaging, which increases the shelf life of development up to 28 days. Still, people are unaware of Vijay dairy's packaging, so we decided to know people about this.
- We made pamphlets and distributed them to the people to use the same strategy for probiotic Dahi awareness.

Problem-3: Distribution of cost for Probiotic Dahi:

How to minimize the cost and give suggestion according to that

- There is already some big brand that has introduced probiotic Dahi. Hence, we suggest reducing costs and reducing the price of Dahi so that customers are attracted to the product in the initial stage.
- Organization use cost sheets as reference documents to help manage purchasing

and production costs, and to find the right selling prices for products and services that's why we help them to classified cost of probiotic dahi. For that we made estimated cost sheet for probiotic dahi.

Problem-4: Shortage of Skilled labour:

How to convince owner need to fulfill shortage of skilled labour

- After the lock down period, the whole industry faced a significant labour shortage. Here we suggested four broad solutions of labour shortage to Vijay Dairy. First, we suggest that Vijay Dairy has to provide travelling expense to their labour so that they come back and start their work. Second, we suggest that Vijay Dairy has to provide leaving place for some month so that they can survive and again settle down. And third is we suggest to give their workers free lunch & meal for some month. The solutions we are suggested to Vijay Dairy are one-time expanse for Dairy but it has many advantages.

IMPLEMENTATION PHASE:

Problem 1: People are not aware about Probiotic Dahi

The aim of this study is to provide a brief overview of consumer's awareness and knowledge towards products containing probiotic in Surat region of Gujarat using a validated questionnaire and is aimed at filling the research gap between consumer's attitude and perceptions towards probiotic dahi and marketing strategies of probiotic dahi. The data was collected with the help of structured questionnaire by simple random sampling. Descriptive statistics was utilized to analyze, conclude and validate the knowledge and attitude of people regarding probiotic dahi. From the data of 300 respondents 69.1% of respondents were familiar with the probiotic dahi out which approximately 36% have ever consumed probiotic. If Vijay dairy introduce probiotic dahi around 87% of people would like to prefer it. Our study indicates that although probiotic market has been established and there is effective marketing strategies and knowledge among consumer attitudes toward probiotic products. We also distributed print ad of brochure of probiotic dahi among approximately 150 customers in Surat city.

Problem 2: People are not aware about Modified Atmospheric Packaging (MAP)

Vijay dairy going to introduced Modified Atmospheric Packaging for that we conduct the survey to provide a brief overview of consumer's awareness and knowledge towards Modified Atmosphere Packaging in Surat region of Gujarat through meeting customer face to face where we explain the knowledge and benefits of MAP. We approximately met 200 customers to spread awareness about modified atmospheric packaging. Before the survey many less people are aware about the MAP but when we met them, we share good knowledge and benefits of modified atmospheric packaging. For the we made print ad brochure to distribute among the people.

Problem 3: Distribution of cost for Probiotic Dahi:

A cost sheet analyzes the components of cost in order to show the per-unit cost for a given product. Organization use cost sheets as reference documents to help manage purchasing and production costs, and to find the right selling prices for products and services that's why we help them to classified cost of probiotic dahi. For that we made estimated cost sheet for probiotic dahi.

Estimated Cost for Probiotic Curd:

For 200gm Probiotic Dahi		
Particular	**Rs.**	**Rs.**
Cost of Raw material	07	
Factory overhead	2.5	
Cost of Production	2.5	
Packaging Cost	01	
Scrap cost	06	
Transportation cost	01	20
Total cost		20
Profit		05
Selling Price		25

Problem 4: Shortage of Skilled labor

As we suggest them to give travelling expanse, they book train tickets for their workers. So mostly all workers come back and they restarted their routine work at Vijay Dairy. They also arrange accommodation place for workers, where workers can stay without paying rent. And lastly, they provide lunch & meal to their workers.

Above shown images are of accommodation place of labors

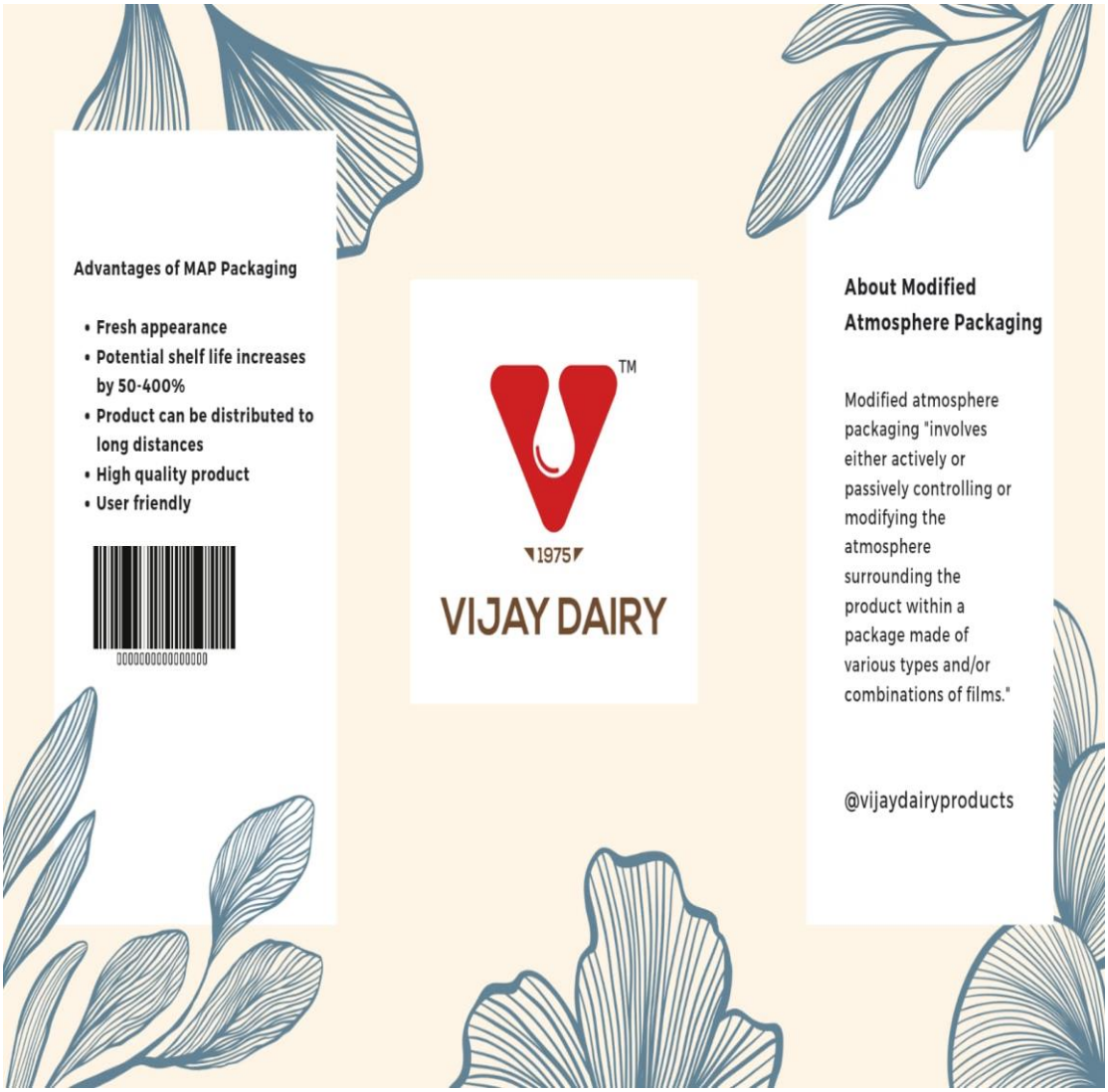
Future Recommendations:

- ✓ Bring out/ launching a new product (i.e. Probiotic Curd) in this health conscious era needs to be of low calories, sugar free, etc.
- ✓ Start to promote as a best/ better substitute to regular curd.
- ✓ Special packaging needs to adopt
- ✓ Introduce multiple flavors within the product line (e.g.- strawberry, Vanilla etc.)
- ✓ Adopt new marketing strategy like Vijay Dairy must need to focus on digital marketing same as in present country delight is doing but again up to some geographical area as it is known in surat only.
  (e.g.- Distributing free samples for trial, Banner & Holdings etc.)
- ✓ Company can also plan to use solar equipment's to minimize the power cost.

ANNEXURE

Brochure for Probiotic Curd

Brochure for Modified Atmosphere Packaging



Questionnaire:

- Which language do you prefer to fill questionnaire for probiotic Dahi?
  (પ્રોબાયોટિક દહીં માટે તમે કઈ ભાષામાં પ્રશ્નાવલી ભરવાનું પસંદ કરો છો?)

  □ English (go to section A)*

  □ ગુજરાતી (go to section B)*

Section A

1. Which Dahi do you prefer Homemade Dahi or Readymade Dahi?

	Not at all Preferred	Least Preferred	Somewhat Preferred	Preferred	Highly Preferred
Homemade Dahi					
Readymade Dahi					

2. Reasons for preferring homemade dahi (If you prefer homemade dahi)*

	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
Homemade Dahi has no preservatives					
It is fresh					
It is less costly					
It has better Quality					

3. When will you prefer to buy a readymade dahi?

	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
If you forget to make dahi at home					
If you have guest at home					
It tastes good than homemade dahi					
It is easily available					
It is available at low price					
It has better quality					

4. Rate your level of awareness about Probiotic Dahi:

Not at all Aware (1)	Least Aware (2)	Somewhat Aware (3)	Aware (4)	Highly Aware (5)

If Rate 4 &5

5. How do you aware of the probiotic Dahi?
   - □ Word of Mouth
   - □ Advertisement
   - □ Company Sales person
   - □ Agents

□ Any other

6. Which dairy brand you prefer to buy probiotic Dahi?

	Not at all Preferred	Least Preferred	Somewhat Preferred	Preferred	Highly Preferred
Amul					
Milky Mist					
Muralya					
Mother Dairy					
Blue Mount Dairy					
Local Dairy					
Homemade					

If Rate 1,2,&3

Description about probiotic Dahi.

Dahi chiefly contains the lactic acid bacteria genus that converts lactose into lactic acid.

A probiotic on the other hand is loaded with a wide variety of beneficial bacteria.

Benefits:

- Improve Digestion
- Contains beneficial bacteria
- Rich in calcium
- Strengthens immune system

7. What factors influenced you to buy this Probiotic Dahi?

	Not at all Preferred	Least Preferred	Somewhat Preferred	Preferred	Highly Preferred
High quality					
Easy availability					
Low price					
Dealer relationship					
Health Concern					
Regular supply					

8. What kind of packaging do you prefer for probiotic Dahi?

	Not at all Preferred	Least Preferred	Somewhat Preferred	Preferred	Highly Preferred
Plastic pouch					
Plastic cup/box					

9. What is the first thing crossing your mind when you see a new brand of probiotic Dahi?

	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree

I have to try it					
Depends on its price.					
Depends if picture on its wrapper looks nice.					
I am not interested in trying it.					
May be, if my friends recommend it.					

10. If Vijay Dairy introduce probiotic Dahi do you like to prefer it?

□ Yes

□ No

11.Name:

_____

12.Gender:

- Female
- Male

13. Age:

- 18-30
- 31-40
- 41-50
- 51-60
- Above 60

14. Area of residence

Zone	Area
North	Amroli, Katargam, Sumul Dairy, Ved Road
East	Bombay Market, Mota Varachha, Nana Varachha, Varachha Road, LH Road, Sarthana
West	Adajan, Rander, Palanpur Patiya, Ramnagar, Anand Mahal Road, Joganinagar, Pal Area, Tadwadi, Ugat Area, L.P Savani Road, Mora Bhagal, Jahangirpura, Hazira, Bhatha, Palanpore, Jahangirabad, Ichchhapore, Ichchhapor, New Cozway Road
South	Udhna, Bamroli, Dindoli, Limbayat, Pandesara, Unn, Parvat Patiya, Udhna Darwaja
Central	Nanpura, Gopipura, Mahidharpura, Rampura, Ring Road, Salabatpura, Begampura, Chowk, Gopi Talav, Nanavat, Rustampura, Sagrampura, Textile Market, Station Road, Sahara Darwaja, Shahpore, Bhagal
South West	Dumas Road, Parle Point, GhodDod Road, Athwalines, Bhatar, City Light, Vesu, Althan, Majura Gate, Piplod, Umra, VN University Road, Green CityLight, Udhna Magdalla Rd, Gauravpath Road, VIP Road, Abhava, Althan Canal Road, Bhimrad

15. Contact No.

_____

Section B

1) તમે કઈ દહીં પસંદ કરો છો હોમમેઇડ દહીં કે રેડીમેડ દહીં?

	બિલકુલ પસંદ	ઓછામાં ઓછું	અંશે	પસંદ	અત્યંત

	નથી	પસંદ	પસંદ		પસંદ
હોમમેઇડ દહીં					
તૈયાર દહીં					

2) હોમમેઇડ દહીંને પ્રાધાન્ય આપવાનાં કારણો (જો તમે ઘરે બનાવેલી દહીં પસંદ કરો છો)*

	ભારપૂર્વક અસંમત	અસંમત	ન સમંત કે ન અસમંત	સંમત	પુરી રીતે સહમત
હોમમેઇડ દહીંમાં કોઈ પ્રિઝર્વેટિવ્સ નથી					
તે તાજુ હોય છે					
તેની કિંમત ઓછી છે					
તેની ગુણવત્તા વધુ સારી છે					

3) તમે ક્યારે તૈયાર દહીં ખરીદવાનું પસંદ કરશો?

	ભારપૂર્વક અસંમત	અસંમત	ન સમંત કે ન અસમંત	સંમત	પુરી રીતે સહમત

જો તમે ઘરે દહીં બનાવવાનું ભૂલી જાઓ છો					
જો તમારા ઘરે મહેમાન છે					
તેનો સ્વાદ ઘરે બનાવેલી દહીં કરતાં સારો છે					
તે સરળતાથી ઉપલબ્ધ છે					
તે ઓછી કિંમતે ઉપલબ્ધ છે					
તેની ગુણવત્તા વધુ સારી છે					

4) પ્રોબાયોટિક દહીં વિશે તમારા જાગૃત સ્તરને રેટ કરો:

બિલકુલ વાકેફ નથી (1)	ઓછામાં ઓછું જાગૃત (2)	કંઈક અંશે જાગૃત (3)	વાકેફ (4)	અત્યંત જાગૃત (5)

જો દર 4 અને 5

5) તમે પ્રોબાયોટિક દહીં વિશે કેવી રીતે જાગૃત છો?

□ વાણી

□ જાહેરાત

□ કંપની સેલ્સ વ્યક્તિ

□ એજન્ટો

□ કોઈપણ અન્ય

6) તમે કઈ ડેરી બ્રાન્ડનું પ્રોબાયોટિક દહીં ખરીદવાનું પસંદ કરો છો?

	બિલકુલ પસંદ નથી	ઓછામાં ઓછું પસંદ	અંશે પસંદ	પસંદ	અત્યંત પસંદ
અમૂલ					
મિલ્કી મિસ્ટ					
મુરલ્યા					
મધર ડેરી					
બ્લુ માઉન્ટ ડેરી					
સ્થાનિક ડેરી					
હોમમેઇડ					

જો દર 1,2,અને 3

પ્રોબાયોટિક દહીં વિશે વર્ણન.

દહીંમાં મુખ્યત્વે લેક્ટિક એસિડ બેક્ટેરિયા જીનસ હોય છે જે લેક્ટોઝને લેક્ટિક એસિડમાં રૂપાંતરિત કરે છે. બીજી તરફ પ્રોબાયોટિક વિવિધ પ્રકારના ફાયદાકારક

બેક્ટેરિયાથી ભરેલું છે.

લાભો:

o પાચનમાં સુધારો

o ફાયદાકારક બેક્ટેરિયા ધરાવે છે

o કેલ્શિયમથી ભરપૂર

o રોગપ્રતિકારક શક્તિને મજબૂત બનાવે છે

7) આ પ્રોબાયોટિક દહીં ખરીદવા માટે તમને કયા પરિબળોએ પ્રભાવિત કર્યો?

	બિલકુલ પસંદ નથી	ઓછામાં ઓછું પસંદ	અંશે પસંદ	પસંદ	અત્યંત પસંદ
ઉચ્ચ ગુણવત્તા					
સરળ ઉપલબ્ધતા					
ઓછી કિંમત					
વેપારી સંબંધ					
આરોગ્યની ચિંતા					
નિયમિત પુરવઠો					

8) પ્રોબાયોટિક દહીં માટે તમે કયા પ્રકારનું પેકેજિંગ પસંદ કરો છો?

	બિલકુલ પસંદ નથી	ઓછામાં ઓછું પસંદ	અંશે પસંદ	પસંદ	અત્યંત પસંદ
પ્લાસ્ટિક બેગ					
પ્લાસ્ટિક કપ/બોક્સ					
પૂંઠું (પેપર બોર્ડ બોક્સ)					

9) જ્યારે તમે પ્રોબાયોટિક દહીંની નવી બ્રાન્ડ જુઓ છો ત્યારે તમારા મગજમાં સૌથી પહેલા શું આવે છે?

	ભારપૂર્વક અસંમત	અસંમત	ન સમંત કે ન અસમંત	સંમત	પુરી રીતે સહમત
મારે તેનો ઉપયોગ કરવો પડશે					
તેની કિંમત પર આધાર રાખે છે.					
તેના રેપર પરનું ચિત્ર સરસ લાગે છે કે કેમ તેના પર					

આધાર રાખે છે.					
મને તેનો ઉપયોગ કરવામાં રસ નથી.					
કદાચ, જો મારા મિત્રો તેની ભલામણ કરે.					

10) જો વિજય ડેરી પ્રોબાયોટિક દહીં રજૂ કરે તો શું તમને તે ખરીદવાનું ગમે છે?

□ હા

□ ના

11) નામ:

_____

12) જાતિ:

○ સ્ત્રી

○ પુરુષ

○ અન્ય

13) ઉંમર:

o 18-30

o 31-40

o 41-50

o 51-60

o 60 થી ઉપર

14) રહેઠાણનો વિસ્તાર

ઝોન	વિસ્તાર
ઉત્તર ઝોન	અમરોલી, કતારગામ, સુમુલ ડેરી, વેડ રોડ
પૂર્વ ઝોન	બોમ્બે માર્કેટ, મોટા વરાછા, નાના વરાછા, વરાછા રોડ, એલએચ રોડ, સરથાણા
પશ્ચિમ ઝોન	અડાજણ, રાંદેર, પાલનપુર પાટિયા, રામનગર, આનંદ મહેલ રોડ, જોગાનીનગર, પાલ વિસ્તાર, તાડવાડી, ઉગત વિસ્તાર, એલ.પી. સવાણી રોડ, મોરા ભાગલ, જહાંગીરપુરા, હજીરા, ભાથા, પાલનપોર, જહાંગીરાબાદ, ઈચ્છાપોર, ન્યુ કોઝવે રોડ.
દક્ષિણ ઝોન	ઉધના, બમરોલી, ડિંડોલી, લિંબાયત, પાંડેસરા, ઉન્ન, પર્વત પાટિયા, ઉધના દરવાજા
સેન્ટ્રલ ઝોન	નાનપુરા, ગોપીપુરા, મહિધરપુરા, રામપુરા, રીંગ રોડ, સલાબતપુરા, બેગમપુરા, ચોક, ગોપી તલાવ, નાણાવત, રૂસ્તમપુરા, સગરામપુરા, ટેક્સટાઈલ માર્કેટ, સ્ટેશન રોડ, સહારા દરવાજા, શાહપોર, ભાગલ

દક્ષિણ પશ્ચિમ ઝોન	ડુમસ રોડ, પાર્લે પોઈન્ટ, ઘોડદોડ રોડ, અઠવાલાઈન્સ, ભટાર, સિટી લાઈટ, વેસુ, અલથાણ, મજુરા ગેટ, પીપલોદ, ઉમરા, વીએન યુનિવર્સિટી રોડ, ગ્રીન સિટીલાઈટ, ઉધના મગદલ્લા રોડ, ગૌરવપથ રોડ, વીઆઈપી રોડ, આભવા, અલથાણ કેનાલ રોડ, ભીમરાડ

15) સંપર્ક નં.

_____